

BS-PH101 — Physics-I

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Module 2 Comprehensive Exam Notes: Optics

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Interference vs. Diffraction and Classification

1.1 Distinction Between Interference and Diffraction

Definition: Interference and Diffraction

- **Interference:** Modification in intensity resulting from the superposition of two or more coherent light waves originating from distinct coherent sources.
- **Diffraction:** The bending of light waves around the sharp edges of an obstacle or aperture and their subsequent superposition within the geometrical shadow region.

Comprehensive Comparison:

Property	Interference	Diffraction
Wave Origin	Superposition of waves from two distinct coherent sources.	Superposition of secondary wavelets originating from different points of the <i>same</i> wavefront.
Fringe Width	Fringes are usually of equal width ($\beta = \lambda D/d$).	Fringes are never of equal width (central max is twice as wide as secondary max).
Intensity	All bright fringes have equal intensity ($I_{\max} = 4I_0$).	Intensity decreases rapidly with increasing order of secondary maxima.
Darkness	Minima are completely dark ($I_{\min} = 0$).	Minima are generally not completely dark.

1.2 Fraunhofer vs. Fresnel Diffraction

Feature	Fresnel Diffraction	Fraunhofer Diffraction
Distance	Source and screen are at finite distances from aperture.	Source and screen are at effectively infinite distances.
Wavefront	Incident wavefront is spherical or cylindrical.	Incident wavefront is plane (parallel rays).
Lenses	No lenses required to focus pattern.	Convex lenses required before and after aperture.
Analysis	Analyzed using Fresnel half-period zones.	Analyzed using simple wave phase summation.

2. Fraunhofer Diffraction at Single, Double, and Multiple Slits

2.1 Single Slit Diffraction

Consider a plane wavefront of wavelength λ incident normally on a slit of width e .

Single Slit Intensity Formula

The resultant intensity on a screen at diffraction angle θ is:

$$I(\theta) = I_0 \left(\frac{\sin \beta}{\beta} \right)^2 \quad \text{where } \beta = \frac{\pi e \sin \theta}{\lambda} \quad (1)$$

Key Conditions:

- **Central Maximum:** At $\theta = 0 \Rightarrow \beta = 0, \lim_{\beta \rightarrow 0} \frac{\sin \beta}{\beta} = 1 \Rightarrow I = I_0$.
- **Minima Condition:** $\sin \beta = 0$ (for $\beta \neq 0$) $\Rightarrow \beta = m\pi \Rightarrow e \sin \theta = m\lambda$ ($m = \pm 1, \pm 2, \pm 3, \dots$).
- **Secondary Maxima:** Occur near $\beta = \pm 1.43\pi, \pm 2.46\pi, \dots$ with intensity ratios:

$$I_0 : I_1 : I_2 = 1 : \frac{1}{22} : \frac{1}{61} = 1 : 0.045 : 0.016 \quad (2)$$

- **Width of Central Maximum:** $\Delta\theta = \frac{2\lambda}{e} \Rightarrow$ Linear Width $W = \frac{2\lambda f}{e}$ (where f is lens focal length).

2.2 Double Slit Diffraction

For two parallel slits, each of width e and separated by opaque spacing d (grating element $= e + d$):

Double Slit Intensity Formula

$$I(\theta) = 4I_0 \left(\frac{\sin \beta}{\beta} \right)^2 \cos^2 \alpha \quad \text{where } \beta = \frac{\pi e \sin \theta}{\lambda}, \quad \alpha = \frac{\pi(e + d) \sin \theta}{\lambda} \quad (3)$$

It is the product of single-slit diffraction factor $\left(\frac{\sin \beta}{\beta} \right)^2$ and interference factor $\cos^2 \alpha$.

Missing Orders (Absent Spectra): Occurs when an interference maximum coincides with a diffraction minimum:

$$(e + d) \sin \theta = n\lambda \quad \text{and} \quad e \sin \theta = m\lambda \Rightarrow \frac{e + d}{e} = \frac{n}{m} \quad (4)$$

If $e = d$, then $\frac{n}{m} = 2 \Rightarrow n = 2m$, so 2nd, 4th, 6th interference orders disappear.

2.3 Diffraction Grating and Resolving Power

A plane transmission diffraction grating consists of N parallel, equally spaced slits with grating element $(e + d)$.

Grating Equations and Resolving Power

- **Principal Maxima Condition:**

$$(e + d) \sin \theta = m\lambda \quad (m = 0, 1, 2, \dots) \quad (5)$$

- **Dispersive Power (D):** Rate of change of diffraction angle with wavelength:

$$D = \frac{d\theta}{d\lambda} = \frac{m}{(e + d) \cos \theta} \quad (6)$$

- **Resolving Power (R):** Capacity to separate two spectral lines of wavelength λ and $\lambda + d\lambda$:

$$R = \frac{\lambda}{d\lambda} = mN \quad (N = \text{total illuminated rulings}) \quad (7)$$

Solved Problem 1: Grating Spectral Lines and Maximum Order

Problem: A grating has 5000 lines/cm. Light of wavelength $\lambda = 6000 \text{ \AA}$ is incident normally. Find: (a) Grating element $(e + d)$, (b) Highest observable order.

Solution:

1. Grating element $(e + d) = \frac{1 \text{ cm}}{5000} = 2 \times 10^{-4} \text{ cm} = 2 \times 10^{-6} \text{ m}$.
2. Maximum angle $\theta_{\max} = 90^\circ \Rightarrow \sin 90^\circ = 1$.
3. Substitute in grating equation $(e + d) \sin 90^\circ = m_{\max} \lambda$:

$$m_{\max} = \frac{(e + d)}{\lambda} = \frac{2 \times 10^{-6} \text{ m}}{600 \times 10^{-9} \text{ m}} = 3.33 \Rightarrow m_{\max} = 3 \quad (8)$$

Hence, maximum 3rd order spectrum can be observed.

Solved Problem 2: Single Slit Central Width

Problem: Light of wavelength $\lambda = 5890 \text{ \AA}$ is incident normally on a slit of width 0.1 mm. A convex lens of focal length $f = 1.0 \text{ m}$ is placed behind the slit. Calculate linear width of central maximum.

Solution:

Linear width of central maximum $W = \frac{2\lambda f}{e}$. $W = \frac{2 \times 5890 \times 10^{-10} \text{ m} \times 1.0 \text{ m}}{0.1 \times 10^{-3} \text{ m}} = 11.78 \times 10^{-3} \text{ m} = 11.78 \text{ mm}$.

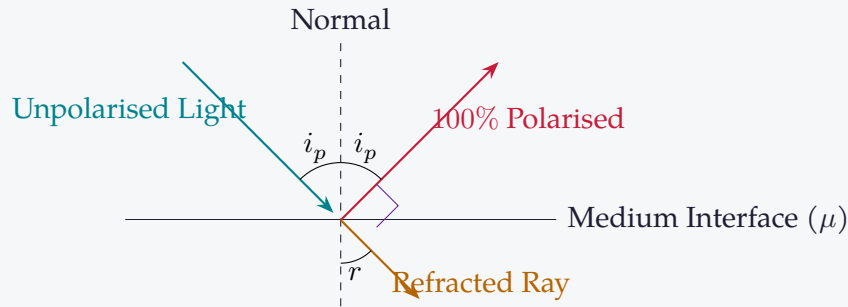
3. Polarisation of Light

3.1 Brewster's Law and Double Refraction

Brewster's Law

When unpolarised light is incident at a specific angle i_p (polarising angle) on a dielectric interface, the reflected light is 100% plane polarised with electric vector perpendicular to plane of incidence.

$$\mu = \tan i_p \quad \text{and} \quad i_p + r = 90^\circ \quad (9)$$

Polarisation by Reflection (Brewster's Law)

Double Refraction (Birefringence): When unpolarised light enters an anisotropic crystal (e.g. Calcite, Quartz), it splits into two plane polarised rays travelling at different speeds:

- **Ordinary Ray (O-ray):** Obeys Snell's law; travels with equal velocity in all directions ($v_o = c/\mu_o$).
- **Extraordinary Ray (E-ray):** Violates Snell's law; velocity depends on direction of propagation (v_e).

3.2 Retardation Plates: Quarter-Wave and Half-Wave Plates**Retardation Plate Thickness Equations**

For a crystal plate cut parallel to optic axis of thickness t :

- **Quarter-Wave Plate (QWP):** Introduces phase difference $\Delta\phi = \pi/2$ or path difference $\Delta x = \lambda/4$:

$$t = \frac{\lambda}{4|\mu_o - \mu_e|} \quad (10)$$

- **Half-Wave Plate (HWP):** Introduces phase difference $\Delta\phi = \pi$ or path difference $\Delta x = \lambda/2$:

$$t = \frac{\lambda}{2|\mu_o - \mu_e|} \quad (11)$$

3.3 Optical Activity and Laurent's Half-Shade Polarimeter

Optically active substances rotate the plane of polarization of linearly polarized light. The angle of rotation θ is given by Biot's law: $\theta = S \cdot L \cdot c$.

$$\text{Specific Rotation } [S]_{\lambda}^T = \frac{\theta}{L \cdot c} \quad \left(\text{in } \frac{\text{deg}}{\text{dm} \cdot \text{g/cm}^3} \right) \quad (12)$$

where θ = rotation angle (deg), L = path length in decimeters (dm), c = concentration in g/cm^3 .

Laurent's Half-Shade Polarimeter: Uses a half-shade plate (half quartz, half glass) to divide the field of view into two halves, enabling precise setting of equal brightness.

Solved Problem 3: Quarter-Wave Plate Thickness

Problem: Calculate the minimum thickness of a quarter-wave plate of Calcite for sodium light of wavelength $\lambda = 5893 \text{ \AA}$. Given $\mu_o = 1.6583$ and $\mu_e = 1.4864$.

Solution:

1. Calcite is a negative crystal ($\mu_o > \mu_e$), so $|\mu_o - \mu_e| = 1.6583 - 1.4864 = 0.1719$.
2. Formula: $t = \frac{\lambda}{4(\mu_o - \mu_e)}$.
3. Substitute: $t = \frac{5893 \times 10^{-10} \text{ m}}{4 \times 0.1719} = \frac{5893 \times 10^{-10}}{0.6876} = 8570 \times 10^{-10} \text{ m} = \mathbf{0.857 \text{ }\mu\text{m}}$.

4. Lasers (Light Amplification by Stimulated Emission of Radiation)

4.1 Principles of Laser Action: Einstein's Coefficients

Interaction of radiation with matter involves three fundamental processes:

Three Radiation Processes and Einstein's Relations

1. **Stimulated Absorption:** Atom absorbs photon: Rate $R_{12} = B_{12}N_1u(\nu)$.
2. **Spontaneous Emission:** Atom decays naturally: Rate $R_{21,\text{sp}} = A_{21}N_2$.
3. **Stimulated Emission:** Incident photon induces decay: Rate $R_{21,\text{st}} = B_{21}N_2u(\nu)$.

At thermal equilibrium, $R_{12} = R_{21,\text{sp}} + R_{21,\text{st}}$. Equating with Planck's law yields Einstein's relations:

$$B_{12} = B_{21} \quad \text{and} \quad \frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3} \quad (13)$$

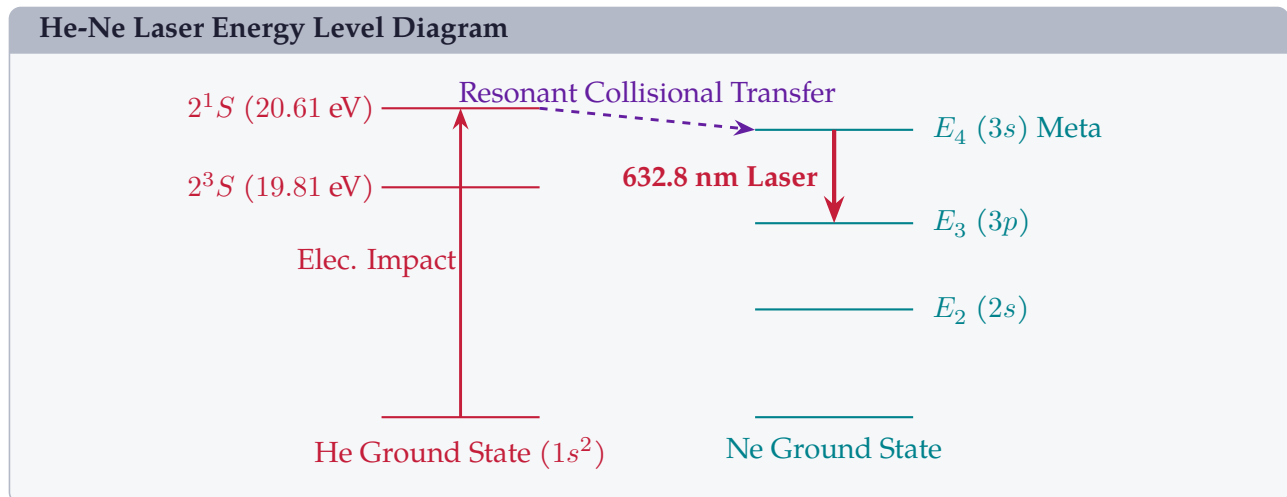
4.2 Population Inversion and Pumping

Under normal thermal equilibrium (Boltzmann distribution $N_2 = N_1 e^{-h\nu/kT}$), $N_1 \gg N_2$.

Definition: Population Inversion and Pumping

- **Population Inversion:** The non-equilibrium state in which higher energy level population exceeds lower energy level ($N_2 > N_1$). Necessary condition for optical amplification.
- **Pumping:** The external process of supplying energy to raise atoms to higher excited levels to achieve population inversion (e.g. Optical pumping, Electrical discharge, Direct electron excitation).
- **Metastable State:** An excited state with a long lifetime ($\sim 10^{-3} \text{ s}$ compared to normal 10^{-8} s), allowing accumulation of atoms for population inversion.

4.3 He-Ne Laser (Gas Laser)



Working Mechanism of He-Ne Laser:

1. Electric discharge accelerates electrons which excite He atoms to metastable 2^1S (20.61 eV) state.
2. He atoms transfer energy to Ne atoms via resonant inelastic collisions, populating Ne's $3s$ (E_4) level.
3. Stimulated emission from $3s \rightarrow 2p$ ($E_4 \rightarrow E_3$) produces red laser light at $\lambda = 632.8$ nm.

Quick Revision — Module 2

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Key Formulas and Metrics at a Glance

Single Slit Minima	$e \sin \theta = m\lambda \quad (m = \pm 1, \pm 2, \dots)$
Single Slit Central Width	$\Delta\theta = \frac{2\lambda}{e} \Rightarrow \text{Linear Width } W = \frac{2\lambda f}{e}$
Double Slit Missing Orders	$\frac{e+d}{e} = \frac{n}{m} \quad (n = \text{interf. order}, m = \text{diff. order})$
Grating Equation	$(e + d) \sin \theta = m\lambda$
Grating Dispersive Power	$D = \frac{d\theta}{d\lambda} = \frac{m}{(e+d) \cos \theta}$
Grating Resolving Power	$R = \frac{\lambda}{d\lambda} = mN$
Brewster's Law	$\mu = \tan i_p \quad \text{and} \quad i_p + r = 90^\circ$
Quarter-Wave Plate (QWP)	$t = \frac{\lambda}{4 \mu_o - \mu_e }$
Half-Wave Plate (HWP)	$t = \frac{\lambda}{2 \mu_o - \mu_e }$
Specific Rotation	$[S]_\lambda^T = \frac{\theta}{L \cdot c} \quad (\text{deg} \cdot \text{dm}^{-1} \cdot (\text{g/cm}^3)^{-1})$
Einstein Coefficients Relation	$\frac{A_{21}}{B_{21}} = \frac{8\pi h \nu^3}{c^3} \quad \text{and} \quad B_{12} = B_{21}$

One-Line Definitions

- **Diffraction Grating:** An optical component with a periodic structure of N slits that diffracts light into several beams.
- **Brewster Angle (i_p):** Angle of incidence at which light with a particular polarization is perfectly transmitted without reflection.
- **Birefringence:** The optical property of a material having a refractive index that depends on the polarization and propagation direction of light.
- **Population Inversion:** State in which a system exists with more members in an excited state than in lower energy states ($N_2 > N_1$).
- **Metastable State:** An excited atomic state with a long lifetime ($\sim 10^{-3}$ s), essential for building up population inversion.

Frequently Asked / Viva Questions

Q1. What is the basic difference between interference and diffraction?

Interference results from superposition of waves from two distinct coherent sources; diffraction results from superposition of wavelets from different parts of the same wavefront.

Frequently Asked / Viva Questions (contd.)

Q2. Why is central maximum in single-slit diffraction twice as wide as secondary maxima?

Because central maximum extends between first minima on either side ($\sin \theta = \pm \lambda/e$), spanning angular width $2\lambda/e$, whereas secondary maxima span width λ/e .

Q3. What is meant by missing orders in double slit diffraction?

Missing orders occur when an interference maximum coincides with a diffraction minimum at the same angle θ , making that order absent ($\frac{e+d}{e} = \frac{n}{m}$).

Q4. State Rayleigh's criterion for spectral resolution.

Two spectral lines are just resolved when the central principal maximum of one falls on the first minimum of the other.

Q5. How does resolving power of a grating differ from dispersive power?

Dispersive power ($d\theta/d\lambda$) measures angular separation between lines; resolving power ($\lambda/d\lambda = mN$) measures ability to distinguish two close wavelengths.

Q6. State Brewster's law and show reflected and refracted rays are perpendicular.

$\mu = \tan i_p$. Since $\frac{\sin i_p}{\sin r} = \tan i_p = \frac{\sin i_p}{\cos i_p} \Rightarrow \sin r = \cos i_p = \sin(90^\circ - i_p) \Rightarrow i_p + r = 90^\circ$.

Q7. What are O-ray and E-ray in a birefringent crystal?

O-ray (Ordinary) obeys Snell's law with constant speed v_o ; E-ray (Extraordinary) violates Snell's law with directional speed v_e .

Q8. Distinguish between positive and negative uniaxial crystals.

In positive crystals ($\mu_e > \mu_o$, e.g. Quartz), $v_o > v_e$; in negative crystals ($\mu_o > \mu_e$, e.g. Calcite), $v_e > v_o$.

Q9. Explain the function of a Quarter-Wave Plate (QWP).

A QWP introduces a path difference of $\lambda/4$ ($\pi/2$ phase shift) between O-ray and E-ray, converting linearly polarized light to circularly or elliptically polarized light.

Q10. What is population inversion and why is it necessary for laser action?

Population inversion ($N_2 > N_1$) ensures that rate of stimulated emission exceeds rate of stimulated absorption, producing net optical amplification.

Q11. Why is Helium gas added to Neon gas in a He-Ne laser?

He atoms are excited by electron impact to metastable states ($2^1S, 2^3S$) and efficiently transfer energy to Ne atoms via resonant collisions, achieving population inversion in Ne.

Q12. Derive relation between Einstein A and B coefficients.

Equating thermal equilibrium rate $B_{12}N_1u(\nu) = A_{21}N_2 + B_{21}N_2u(\nu)$ with Boltzmann distribution $N_2/N_1 = e^{-h\nu/kT}$ and Planck's radiation law gives $B_{12} = B_{21}$ and $A_{21}/B_{21} = 8\pi h\nu^3/c^3$.

Summary Comparison of Laser Processes

Process	Incident Condition	Transition	Output Properties
Absorption	Photon ($h\nu = E_2 - E_1$)	Lower $E_1 \rightarrow$ Higher E_2	Atom reaches excited state.
Spontaneous Emission	Natural decay (lifetime $\sim 10^{-8}\text{s}$)	Higher $E_2 \rightarrow$ Lower E_1	Incoherent, random phase/direction.
Stimulated Emission	Triggering photon ($h\nu$)	Higher $E_2 \rightarrow$ Lower E_1	Coherent, identical phase, direction, and wavelength.